Master Business Requirement Document & Technical Specification (BRD)

Project Title: Shri Khatu Shyam Temple Digital Platform

Project Lead: Ayush Sureka

Target Location: Dalsinghsarai, Samastipur (Bihar)

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1. Executive Summary & Project Vision

The objective of this project is to build a professional, secure, and highly scalable digital platform for the Shri Khatu Shyam Temple in Dalsinghsarai, Bihar. The platform bridges the physical gap between the temple and devotees worldwide, enabling seamless online ritual bookings, flexible contributions (via automated payment gateway or free UPI QR code), and instant 80G tax-compliant receipt generation.

Designed for long-term reliability and operational transparency, the system eliminates manual paperwork, prevents festival overbooking through automated capacity checks, and provides devotees with a 24/7 self-service dashboard.

2. Target Audience & Role-Based Access Levels

The platform follows a clean access model: **General browsing requires no account or login.** Users only register or log in when they want to take a personal action requiring their identity (such as booking a ritual, donating, or viewing their personal history).

User Role	Access & Permissions	Login Required?

Anonymous Visitor (General Public)	• Explore the homepage, About section, and temple history. • View upcoming events, festival dates, and timings. • View address, contact details, embedded Google Map, and social links. • Use the instant English/Hindi bilingual language toggle.	**No** (Open to all)

Registered Devotee (Member)	• Submit Aarti and Bhog ritual bookings. • Make secure online contributions. • Access the **Devotee Dashboard** to track past/upcoming visits and re-download official tax receipts instantly.	**Yes** (Secure login session via JWT)
Temple Administrator (Staff)	• Full backend management access. • Add, update, or remove dynamic event schedules.	**Yes** (Secure admin authentication)

	• Monitor financial collection reports and ritual booking capacities.	

3. Core Website Modules & Features

The website is structured into intuitive sections designed for optimal devotee engagement:

- **Hero & Bilingual Header:** Features traditional temple branding, a welcoming banner (*"Hare Ka Sahara, Baba Shyam Hamara"*), and an instant **English/Hindi language switcher** powered by a client-side JavaScript dictionary that translates the entire page instantly without reloading.
- **About The Temple:** Highlights the spiritual significance of Barbarika (Shyam Baba) and offers guidance for pilgrims visiting the region.
- **Dynamic Event Calendar:** Displays upcoming festivals, Ekadashi darshans, and special temple events clearly featuring the **Event Name (Bilingual), Exact Date, and Time**, automatically sorted chronologically.
- **Ritual & Booking Portal:** An interactive form allowing devotees to book Aarti slots or Bhog offerings. Selecting a category dynamically updates the specific ritual options, capturing the devotee's name, gotra, date, and number of participants.
- **Online Donation & Tax Compliance:** Secure contribution forms mandating the donor's name, PAN number (for tax records), and amount.
- **Location, Map & Contact:** Complete physical address details for Dalsinghsarai, travel directions (nearest railway and airport), and an embedded interactive Google Map.
- **Social Media & Direct Support:** Direct links to the temple's official Facebook, Instagram, and YouTube channels, plus a **floating WhatsApp widget** for instant devotee inquiries.
- **Media Gallery:** High-resolution imagery of the deity, sanctum, and festival celebrations.

4. Devotee Dashboard & Self-Service Portal

Once logged in, registered devotees gain access to a dedicated account hub featuring:

1. **Bookings Tracker:** Clear division between *Upcoming Bookings* (confirmed slots with ticket download options) and *Past Bookings* (archived ritual history).

2.
3. **Payment & Donation History:** A transparent ledger of all past contributions showing transaction IDs, dates, amounts, and payment status.
4.
5. **Instant Receipt Re-Download:** A dedicated **"Download 80G Receipt"** button next to every successful donation, allowing devotees to pull fresh PDF copies of their tax certificates on demand without contacting temple administration.
6.

5. Automated Notifications & Compliance

- **Instant Email Dispatch:** Upon successful payment or booking confirmation, the system automatically generates an official, tamper-proof PDF receipt in memory (via OpenPDF) and emails it directly to the devotee's inbox using a professional custom domain email (info@khatushyamtemple.in) via Brevo SMTP.
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- **Tax Compliance:** Receipts are formatted to meet standard religious trust compliance guidelines, displaying necessary trust details and PAN data for income tax exemption purposes.
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6. Hybrid Payment Architecture (Gateway + Free UPI QR)

To balance absolute zero-cost options with automated convenience, the platform supports two parallel payment channels:

1. **Direct UPI QR Code (100% Free):**
2.
    - **How it works:** Displays the Temple Trust's official static UPI QR code on screen. Devotees scan it using any app (Google Pay, PhonePe, Paytm) to transfer funds directly into the bank account with **₹0 commission or gateway fees**.
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    - *Trade-off:* Requires manual verification of the transaction reference number by the user or admin for automated receipt generation.
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3. **Payment Gateway (Razorpay / PayU):**
4.
    - **How it works:** Automatically handles Credit Cards, Debit Cards, Netbanking, NEFT, and automated UPI. Instantly verifies payment via secure webhooks to trigger instant 80G PDF generation.
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    - *Trade-off:* Incurs a standard processing fee (~2.36% inclusive of GST) per transaction, with automatic settlements directly into the Trust's bank account.
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7. Developer-Centric Engineering Specifications

To ensure absolute clarity for software developers taking over the implementation, the backend and frontend adhere to strict architectural patterns:

A. Tech Stack & Software Requirements

- **Backend:** Java 17, Spring Boot 3.x (Web, Data JPA, Security).
- **Database:** MySQL Server 8.0+ with **Flyway / Liquibase** for automated database schema migrations stored in `src/main/resources/db/migration`.
- **Frontend:** HTML5, CSS3, Vanilla JavaScript (ES6+).
- **Server / Proxy:** Nginx reverse proxy routing traffic from port 443 (HTTPS) to Spring Boot on port 8080.
- **Documentation: Springdoc OpenAPI (Swagger UI)** auto-generated at `/swagger-ui/index.html`.

B. Authentication & Authorization (Security Layer)

- **Stateless JWT:** Users log in (`POST /api/auth/login`), receive a signed JWT, and store it in `localStorage`. Subsequent requests attach the token via `Authorization: Bearer <token>`. Sessions are strictly stateless (`SessionCreationPolicy.STATELESS`).
- **Password Security:** BCrypt hashing for all user and admin passwords.
- **CORS Configuration:** Global `WebConfig` explicitly permitting communication between the frontend domain and the backend API.

C. Error Handling & Logging

- **Global Exception Handler:** Implemented via `@ControllerAdvice` to return clean JSON error payloads (e.g., `{"status": 400, "message": "Slot capacity reached"}`) instead of raw stack traces.
- **Logging:** SLF4J / Logback configured to record critical booking and payment errors into dedicated log files (`/var/log/temple/app-errors.log`).

8. Infrastructure, Security & Disaster Recovery

To ensure the platform runs 24/7 without crashing during high-traffic periods (like Phalgun Mela) and protects sensitive financial records, it utilizes a professional production architecture:

- **Cloud Server (VPS):** Hosted on a dedicated Virtual Private Server running Ubuntu Linux with a **minimum of 2GB RAM** to smoothly handle backend Java operations and database queries.

- **Web Server & Security:** Configured using Nginx as a secure reverse proxy paired with an auto-renewing **SSL Certificate (Let's Encrypt / Certbot)** for encrypted data transmission.
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- **Off-Site Automated Backups:** A scheduled daily Bash script (`backup_db.sh`) runs via Linux `cron` at 2:00 AM, dumping and compressing the MySQL database before syncing an encrypted copy to an isolated off-site cloud storage location (Cloudflare R2 or Google Drive), ensuring zero data loss.
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9. Financial Summary & Cost Breakdown

The financial structure is optimized for long-term sustainability with minimal fixed overhead:

Expense Category	Item Description	Estimated Cost (INR)
Fixed Annual Outlay	Domain Registration (`.in` or `.com`)	₹700 – ₹1,000 / year
Fixed Annual Outlay	Cloud Server (2GB RAM Ubuntu VPS)	~₹5,500 – ₹7,000 / year
Fixed Annual Outlay	Off-site Backups & SSL Security	₹0 (Using free cloud tiers & Let's Encrypt)
Fixed Annual Outlay	Professional Email & SMTP Delivery	₹0 (Using Brevo free transactional tier)
Total Fixed Cost	**Annual Run Cost to keep portal live 24/7**	**~₹6,500 – ₹8,000 / year**

Variable Cost	Payment Gateway Processing Fee	~2.36% per transaction (Only paid when using automated gateway; **₹0 when using Free UPI QR**)

10. Legal Compliance & Trust Prerequisites

Before the digital platform can go live and legally collect donations or issue 80G tax-exemption receipts, the client (Temple Trust) must fulfill specific regulatory requirements:

- **Trust/Society Registration:** The temple must be formally registered as a legal entity under Indian law (Trust Deed, Society Registration Act, or Section 8 Company). Donations cannot legally be funneled into a personal savings account.
- **PAN & Current Bank Account:** A dedicated PAN card must be issued in the name of the Trust, and a Current Bank Account must be opened to receive automated Razorpay settlements or direct UPI transfers.
- **80G & 12A Certification:** Official certification from the Income Tax Department enabling the trust to provide tax-deductible receipts to donors.

11. Development Milestones & Delivery Timeline

A structured timeline ensures clear accountability and milestone tracking during the build phase:

- **Phase 1 (Setup & Database Design - Week 1):** Finalizing database schema (`temple_db`), configuring Flyway migrations, setting up local development environments (IntelliJ, MySQL), and setting up the Git code repository.
- **Phase 2 (Core Development - Weeks 2-3):** Building backend Spring Boot REST APIs, implementing JWT authentication, setting up user roles (User/Admin), and writing core booking and donation logic with `@Transactional` safety.
- **Phase 3 (Frontend Integration - Week 4):** Developing the bilingual HTML/CSS/JS frontend, connecting UI forms to backend APIs, and implementing dynamic event feeds and the language switcher.
- **Phase 4 (Payment & Notifications - Week 5):** Integrating Razorpay webhooks, signature verification, OpenPDF receipt generation, Brevo SMTP email dispatch, and the static UPI QR option.
- **Phase 5 (Server Deployment & Testing - Week 6):** Provisioning the 2GB Ubuntu VPS, configuring Nginx, SSL certificates, systemd services, and automated off-site cloud backups. Go-live execution.